Valuing Minnesota Family Forests: Economic and Ecological Contributions*

Ryan McWay

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Abstract

This report assesses the monetary and non-monetary value of Minnesota family forests, with particular attention to the ecosystem services they provide. First, it establishes a framework for evaluating these services in both ecological and economic terms. Then, for a selection of high priority ecosystem services I estimate the economic value attributable to Minnesota forests. I demonstrate heterogeneity in value across forest conditions, ownership types, and across the time horizion 2003 - 2023. This report therefore provides both a conceptual framework and empirical analysis for understanding the value of Minnesota family forests.

Keywords: Family Forests; Ecosystem Services; Forest Valuation; Minnesota

1 Introduction

Describe the importance of Minnesota family forests, the decision context for this assessment, and the research questions addressed by the report. - Place within the strategic plan of MFRC. - Is there a broader (national) context to place this within.

2 The State of North America's Forests

Review the relevant literature, regional context, and ecosystem-service concepts that guide the analysis.

- Description of NA, national, and MN forests - Narrowing to family forests

2.1 Minnesota Family Forests

3 Methodology to Value Minnesota Family Forests

3.1 Study Scope

Define the study area, forest-owner population, ecosystem services, and time period included in the assessment.

*This project recieved considerable support from David Wilson and Amanda Hoffman.

3.2 Forest Ecosystems Assessed

- Board overview of ecosystem services - Specific ecosystem services evaluated - within the context of Gross-ecosystem product (GEP) value and natural capital accounting - This is not a strict accounting approach, but will attempt to be congruent with SEEA-EA accounting framework.

3.3 List of Potential Ecosystem Services attributable to Forests

Ecosystem Service	Forest Mechanism	Feasibility	Feasibility: Quantity	Feasibility: Price	Feasibility: Lambda	Category	Direct/Indirect
Timber Provision	Wood products	High	MN FIA forest raster; FIA species, DBH, height, merchantable volume and growth; cell area; ownership and access overlays	MN stumpage prices by species/grade; harvest and haul costs	Net timber rent: $\sum_i A_i Q_i^{sust}(P_i - C_i)$	Provisioning	Direct
Carbon Sequestration	photosynthetic uptake stored in biomass and soils	High	MN FIA forest raster; repeated FIA carbon stocks, growth, mortality and removals; cell area	Social cost of carbon or carbon price per tCO2e	Annual flow: $\sum_i A_i[(C_{i,t2} - C_{i,t1})/\Delta t](44/12)P_{CO2}$	Regulating	Indirect
Carbon Storage	carbon held in live biomass, deadwood, litter, and soil	High	MN FIA forest raster; carbon stock by pool (tC/ha); cell area	Social cost of carbon per tCO2e	Stock value: $\sum_i A_i C_i(44/12)P_{SCC}$; $\lambda = 1$	Regulating	Indirect
Habitat Support for biodiversity	Forest composition and structure support species and habitat	Medium	MN FIA forest raster; FIA species composition and stand structure; species range/occurrence data; cell area	Species-specific WTP or restoration cost per quality-adjusted ha	$V = \sum_i A_i H_i u_i \lambda_i$	Supporting	Indirect (intermediate)

Ecosystem Service	Forest Mechanism	Feasibility	Feasibility: Quantity	Feasibility: Price	Feasibility: Lambda	Category	Direct/Indirect
Recreation: Hunting	Forests provide game habitat and hunting settings	High	MN FIA forest raster; hunter-days/trips by location and game; licenses and participation survey	Consumer surplus per trip/day from travel-cost or stated-preference estimates	$V = \sum_j D_j CS_j \lambda_j$; forest-attributable share	Cultural	Direct
Recreation: Fishing	Forested watersheds support fish habitat and water quality	High	MN FIA forest raster; lake/river watersheds; angler-days/trips and fishery data	Consumer surplus per trip/day from travel-cost estimates	$V = \sum_j D_j CS_j \lambda_j$; forest-watershed-attributable share	Cultural	Direct
Recreation: Nature recreation	Forest settings support day-use hiking, camping, wildlife viewing, paddling and biking	High	MN FIA forest raster; visits/user-days by forested site; trail counters and permits	Consumer surplus per visit from travel-cost demand model	$V = \sum_j D_j CS_j \lambda_j$; forest-attribute share	Cultural	Direct
Recreation: Nature tourism	Forest landscapes draw overnight and long-distance visitors	High	MN FIA forest raster; overnight trips, visitor-days and lodging/campground nights by destination	Consumer surplus per trip from travel-cost demand model (including transport and lodging costs)	$V = \sum_j T_j CS_j \lambda_j$; forest-purpose share	Cultural	Direct

Ecosystem Service	Forest Mechanism	Feasibility	Feasibility: Quantity	Feasibility: Price	Feasibility: Lambda	Category	Direct/Indirect
Recreation: Winter and motorized	Trails for snowmobiling, skiing, ATV, snowshoeing	High	Trail miles or user days (registrations, permits)	Travel cost; trail pass and registration fees	Share of trip value due to forest setting vs. equipment and trail grooming	Cultural	Direct
Habitat Support for biodiversity	Structurally diverse forest supporting species richness	Medium	habitat acres and species richness	Species-specific WTP or replacement/restoration cost	Share of habitat value attributable to forest type	Supporting	Indirect (intermediate)
Water Quality Regulation (Riparian Buffers)	Riparian forest buffers retain nitrogen and phosphorus before delivery to streams	Medium	InVEST NDR nutrient export with buffers vs. no-buffer counterfactual; kg N and P per year by watershed	Avoided treatment/damage cost or nutrient-credit price per kg N/P	$\Delta E_{\text{family forest}}/\Delta E_{\text{all buffers}}$	Regulating	Indirect
Air quality regulation	Canopy removes particulate matter and gaseous pollutants	Medium	MN FIA forest raster; canopy/leaf area and pollutant removal by species; exposed population	Marginal health damages avoided per ton removed	$V = \sum_{i,p} R_{ip} MD_{ip} \lambda_{ip}$; canopy-attributable share	Regulating	Indirect
Local climate regulation	Shade and evapotranspiration cooling; windbreaks	Medium	Canopy cover; degrees of cooling; reduced heating/cooling kWh	Avoided energy cost	Share of energy savings attributable to trees vs. building characteristics	Regulating	Indirect

Ecosystem Service	Forest Mechanism	Feasibility	Feasibility: Quantity	Feasibility: Price	Feasibility: Lambda	Category	Direct/Indirect
Erosion control and soil retention	Roots and litter stabilize slopes and banks	Medium	Tons of soil retained (RUSLE, forest vs. counterfactual cover)	Replacement cost (dredging, lost productivity)	Forest-attributable share vs. other cover types	Regulating	Indirect
Woody biomass energy	Mill residues, harvest slash, pellets	Medium	Dry tons of biomass	Delivered biomass or pellet price	Strip harvest and processing labor and capital	Provisioning	Direct
Property value amenity	Proximity to or views of forest	Medium	Homes within a set distance of forest	Hedonic price premium	Ratio of premium to forest attribute (acres, canopy %)	Cultural	Direct (capitalized into land value)
Physical and mental health	Reduced stress, physical activity, lower heat and air pollution exposure	Medium	Visits or green-space exposure	Avoided health-care costs; QALY or VSL-based	Dose-response attributable to forest vs. other green space	Cultural	Indirect
Non-timber forest products (NTFP)	Berries, nuts, syrups, mushrooms, root extracts, cosmetic extracts	Medium	harvest (permits, gathering surveys)	market prices	production function residual (resource rent)	Provisioning	Direct
Nature disaster prevention	Healthy forests have erosion prevention to mitigate flooding, drought, storm impacts	Medium	acres forested floodplain, flood/wildfire frequency	Avoided damages (FEMA claims, property loss)	Forest share of risk reduction vs. engineered defenses	Regulating	Indirect

Ecosystem Service	Forest Mechanism	Feasibility	Feasibility: Quantity	Feasibility: Price	Feasibility: Lambda	Category	Direct/Indirect
Water Supply	Infiltration, groundwater recharge, baseflow regulation	Low	Cubic meters of yield from forested watersheds	utility water rates less treatment and distribution costs	residual after removing treatment and capital/labor	Provisioning	Direct
Fuelwood	Subsistence energy production via deadwood collection	Low	Cords gathered	stumpage cordwood price	Regress out labor and transport to recover residual.	Provisioning	Direct
Soil capitalization	soil formation via agroforestry services	Low	SOC accumulated, topsoil retained	avoided erosion damage, fertilizer cost	forest attributable share	Supporting	Indirect
Nature education	Forested sites serve as outdoor classrooms for environmental learning (school field trips, scouting, nature centers, interpretive trails, forest schools, field stations)	Low	Scouting activities or presence of nature education programs in forests	Travel cost	Infeasible. What is attributable to the forest vs. what is human capital/knowledge transfer.	Cultural	Indirect
Cultural value of identity	Forests as a sense of identity provided to local populations	Low	PPIG survey attachment measures	Stated WTP preferences	Infeasible.	Cultural	Indirect

Ecosystem Service	Forest Mechanism	Feasibility	Feasibility: Quantity	Feasibility: Price	Feasibility: Lambda	Category	Direct/Indirect
Pollination	Native bee and insect habitat that supports adjacent crops	Low	Crop acres within foraging range of forest	Managed hive rental rates or crop yield uplift	Share of yield uplift due to wild vs. managed pollinators	Regulating	Indirect (intermediate)
Pest and disease regulation	Predator habitat (birds, bats, insects)	Low	Reduced pest outbreak frequency or acres	Avoided pesticide and crop-loss costs	Forest-attributable share	Regulating	Indirect (intermediate)
Nutrient cycling and primary production	Litterfall, decomposition, biomass growth	Low	Net primary productivity	Not separately priced (intermediate input to timber, carbon, soil)	Infeasible, since this would double-count	Supporting	Indirect (intermediate)
Genetic resources / bioprospecting	Genetic diversity of plants, fungi, microbes	Low	Not well-defined	Option value or pharma royalties	Infeasible	Provisioning	Direct (option value)
Existence and bequest value	Satisfaction from knowing forests persist	Low	Households with positive WTP	Contingent valuation or choice experiments	Hard to separate from other values; use WTP per attribute	Cultural	Indirect (non-use)
Tribal and Indigenous cultural uses	Maple sugaring, ceremonial and subsistence harvest, treaty rights	Low	Participation and harvest levels (tribal data, with consent)	Mostly non-market; avoid imposing market prices	Infeasible and arguably inappropriate to monetize	Cultural	Direct and indirect

Scientific research	Long-term research sites and experimental forests	Low	Research funding or publications tied to the site	Research expenditure (cost-based)	Infeasible	Cultural	Indirect

3.3.1 Ecosystem 1: Timber Provision

Data. Use the Minnesota FIA forest raster as the spatial base, with each forest pixel i having area A_i in hectares. Attach FIA estimates of species, diameter, height, merchantable volume, annual growth, mortality, and removals to the forest pixels. Add forest-ownership data if the estimate is restricted to family forests, and road, mill, and species/grade-specific Minnesota stumpage-price data to estimate access and costs. The forest raster identifies the analysis footprint; FIA inventory attributes supply the forest measurements.

Valuation Approach. For pixel i, calculate annual sustainable merchantable harvest quantity from FIA growth, mortality, removals, and the selected harvest/retention rule:

$$Q_i = A_i q_i^{\text{sust}},$$

where A_i is pixel area and q_i^{sust} is sustainable merchantable volume per hectare per year. Calculate the net stumpage price:

$$P_i = p_{i,\text{stumpage}} - c_{i,\text{harvest}} - c_{i,\text{haul}},$$

using species- and grade-specific Minnesota prices and variable costs.

Currently, set nature's contribution to be fully attributable to the eligible family forest:

$$\lambda_i = 1.$$

Then calculate

$$V_{\text{timber}} = \sum_i Q_i P_i \lambda_i,$$

with Q_i and P_i in matching volume and currency units. Report the quantity and value period consistently; do not value the full standing inventory as an annual flow – requires assumptions about rotation length and sustainable yield.

Estimated Value

3.3.2 Ecosystem 2: Carbon Sequestration

Data. Use the Minnesota FIA forest raster as the spatial base. Estimate change in carbon stocks per hectare from remeasured FIA plots or FIA annual estimates, accounting for growth, mortality, and removals over the measurement interval. Record the included carbon pools, inventory years, and cell area A_i; use a consistent pool definition across inventories. Obtain a social cost of carbon or carbon price in dollars per tCO2e.

Valuation Approach. For pixel i, calculate annual net sequestration from repeated FIA carbon inventories, multiplying by pixel area and converting tC to tCO_2e:

$$Q_i = A_i \left(\frac{C_{i,t_2} - C_{i,t_1}}{t_2 - t_1} \right) \left(\frac{44}{12} \right),$$

where the same carbon pools are included in both inventories. $\left(\frac{44}{12}\right)$ is molecular-weight conversion from carbon (C) to carbon dioxide (CO_2). Carbon has an atomic mass of about 12. CO_2 has a molecular mass of about $44 : 12$ for carbon $+16\times2$ for oxygen. So 1 tonne of carbon $= 44/12 \approx 3.667$

tonnes of CO_2. Select a social cost of carbon or carbon-market price, stated in a consistent dollar year:

$$P_i = P_{\mathrm{CO}_2} \quad [\$/\mathrm{tCO_2e}].$$

Currently, set nature's contribution to be fully attributable to the eligible family forest:

$$\lambda_i = 1.$$

Value the annual flow as

$$V_{\text{sequestration}} = \sum_i Q_i P_i \lambda_i,$$

where Q_i is annual net $\mathrm{tCO_2e}$ and P_i is dollars per $\mathrm{tCO_2e}$. The result is an annual flow value; do not add the full carbon-stock value to it without defining a non-overlapping valuation period.

Estimated Value

3.3.3 Ecosystem 3: Carbon Storage

Data. The spatial base is the Minnesota FIA forest raster, with pixel area A_i. Required attributes are FIA carbon stocks per hectare by pool (live trees, deadwood, litter, and soil where available) and the pool coverage and inventory date. Convert carbon to CO2-equivalent using 44/12.

Valuation Approach. For pixel i, calculate the total carbon stock in the included pools, multiplied by pixel area and converted from tC to $\mathrm{tCO_2e}$:

$$Q_i = A_i C_i \left(\frac{44}{12}\right),$$

where C_i is the combined pool stock in tC/ha. Select a social cost of carbon and state its dollar year:

$$P_i = P_{\mathrm{SCC}} \quad [\$/\mathrm{tCO_2e}].$$

Currently, set nature's contribution to be fully attributable to the eligible family forest:

$$\lambda_i = 1.$$

Value the stock as

$$V_{\text{storage}} = \sum_i Q_i P_i \lambda_i,$$

where Q_i is $\mathrm{tCO_2e}$ and P_i is dollars per $\mathrm{tCO_2e}$. This is a one-time stock value, not an annual flow; keep it separate from sequestration value to avoid double counting.

Estimated Value

3.3.4 Ecosystem 4: Habitat Support for Biodiversity

Data. Use the Minnesota FIA forest raster as the spatial base. Required forest attributes include FIA species composition, tree size and basal area, and available structural measures such as snags and down wood; add species range or occurrence data for the selected taxonomic groups. Derive a species-specific habitat suitability score H_i for each pixel using a stated habitat model, then multiply by cell area A_i. Obtain a species-specific willingness-to-pay or restoration-cost estimate u_i per quality-adjusted hectare and estimate the forest-attributable share λ_i.

Valuation Approach. Calculate quality-adjusted habitat area for species s and pixel i:

$$Q_{is} = A_i H_{is},$$

where H_{is} is the habitat-suitability score from FIA composition/structure and species range or occurrence data. Select a species-specific willingness-to-pay or restoration-cost estimate per quality-adjusted hectare:

$$P_s = u_s \quad [\$/\text{quality-adjusted ha}].$$

Estimate the fraction of suitable habitat attributable to forest by comparing the modeled forest condition with its nonforest counterfactual:

$$\lambda_{is} = \frac{\max(0, H_{is}^{\text{forest}} - H_{is}^{\text{counterfactual}})}{H_{is}^{\text{forest}}}, \quad H_{is}^{\text{forest}} > 0.$$

Then value habitat as

$$V_{\text{habitat}} = \sum_{i,s} Q_{is} P_s \lambda_{is}.$$

Keep the valuation basis consistent across species and avoid summing overlapping values for the same habitat. If using restoration costs, report them as a cost-based proxy rather than as willingness to pay.

Estimated Value

3.3.5 Ecosystem 5: Recreation—Hunting

Data. Use the Minnesota FIA forest raster as the spatial base and summarize forest cover, composition, and structure around hunting destinations or within game-species ranges. Required additional data include hunter-days or trips by location and game species, license and participation survey data, and relevant game abundance or harvest estimates. Obtain consumer-surplus estimates per hunter-day or trip from Minnesota-specific or comparable travel-cost or stated-preference studies. Estimate λ_j, the share of hunting value attributable to forest habitat and setting rather than game abundance, access, regulations, equipment, or other trip features.

Valuation Approach. Estimate annual hunter-days or trips for destination/game area j from weighted participation surveys and administrative counts:

$$Q_j = \sum_{r \in j} w_r d_r,$$

where d_r is reported hunter-days or trips and w_r is the survey expansion weight. Estimate consumer surplus per matching unit with a travel-cost demand model or stated-preference study:

$$P_j = CS_j \quad [\$/\text{hunter-day}] \text{ or } [\$/\text{trip}].$$

Estimate the share of consumer surplus attributable to forest habitat from a forest-versus-counterfactual model, controlling for game abundance, access, regulations, equipment, and other trip features:

$$\lambda_j = \frac{\max(0, CS_j^{\text{forest}} - CS_j^{\text{counterfactual}})}{CS_j^{\text{forest}}}, \quad CS_j^{\text{forest}} > 0.$$

Then value hunting as

$$V_{\text{hunting}} = \sum_j Q_j P_j \lambda_j,$$

using a consistent time period and avoiding addition of license revenue or trip expenditures to consumer surplus.

Estimated Value

3.3.6 Ecosystem 6: Recreation—Fishing

Data. Use the Minnesota FIA forest raster to measure forest cover and composition within lake and river watersheds and near fishing access points. Combine it with angler-days or trips by waterbody, creel or participation surveys, fishery characteristics, and water-quality data. Obtain consumer surplus per angler-day or trip from a travel-cost demand model or a suitable Minnesota study. Estimate λ_j, the portion of fishing value attributable to forest-related watershed conditions (such as shade, runoff, and water quality), rather than stocking, access, regulations, or other factors.

Valuation Approach. Estimate annual angler-days or trips at waterbody/destination j from weighted creel/participation surveys and administrative counts:

$$Q_j = \sum_{r \in j} w_r d_r,$$

where d_r is reported angler-days or trips and w_r is the survey expansion weight. Estimate consumer surplus per matching unit with a travel-cost demand model:

$$P_j = CS_j \quad [\$/\text{angler-day}] \text{ or } [\$/\text{trip}].$$

Estimate the forest-watershed-attributable share of surplus by comparing modeled forest conditions with the watershed counterfactual, controlling for stocking, access, regulations, and other factors:

$$\lambda_j = \frac{\max(0, CS_j^{\text{forest watershed}} - CS_j^{\text{counterfactual}})}{CS_j^{\text{forest watershed}}}, \quad CS_j^{\text{forest watershed}} > 0.$$

Then value fishing as

$$V_{\text{fishing}} = \sum_j Q_j P_j \lambda_j,$$

avoiding addition of trip expenditures to consumer surplus.

Estimated Value

3.3.7 Ecosystem 7: Recreation—Nature Recreation

Data. Use the Minnesota FIA forest raster to characterize forest cover and attributes at forested public lands, trails, and other day-use destinations. Required visitation data include annual visits or user-days by destination, from trail counters, permits, or visitor surveys. Obtain travel costs and visitation choices (including round-trip distance, vehicle cost, and travel time) to estimate consumer surplus per visit using a travel-cost demand model. Estimate λ_j, the share of recreation value attributable to forest attributes, distinct from trail facilities and other amenities.

Valuation Approach. Estimate annual visits to day-use destination j from counters, permits, or weighted visitor surveys, correcting for uncounted visits:

$$Q_j = \frac{N_j^{\text{observed}}}{c_j},$$

where c_j is estimated visitor-count coverage (use $c_j = 1$ for a complete count). Estimate consumer surplus per visit using a travel-cost demand model with trip distance, vehicle cost, and travel time:

$$P_j = CS_j \quad [\$/\text{visit}].$$

Estimate the share of consumer surplus attributable to forest attributes by comparing forested-site surplus with a counterfactual that retains trails and other nonforest amenities:

$$\lambda_j = \frac{\max(0, CS_j^{\text{forest site}} - CS_j^{\text{counterfactual}})}{CS_j^{\text{forest site}}}, \quad CS_j^{\text{forest site}} > 0.$$

Then value nature recreation as

$$V_{\text{nature recreation}} = \sum_j Q_j P_j \lambda_j,$$

using the travel-cost model to estimate welfare rather than treating visitor spending as welfare, and avoiding double counting visits across sites.

Estimated Value

3.3.8 Ecosystem 8: Recreation—Nature Tourism

Data. Use the Minnesota FIA forest raster to characterize forest cover and attributes at overnight and long-distance destinations. Assemble annual visitor-trips or visitor-days by destination from tourism surveys, campground reservations, lodging data, and visitor counts. Collect trip origins, transport and lodging costs, and trip purpose to estimate consumer surplus per trip from a travel-cost demand model. Estimate λ_j, the forest-attributable share of trip value, using trip-purpose or stated-preference evidence.

Valuation Approach. Estimate annual overnight or long-distance trips to destination j from weighted tourism surveys, lodging/campground records, and visitor counts, using a consistent trip definition:

$$Q_j = \frac{T_j^{\text{observed}}}{c_j},$$

where c_j is estimated trip-count coverage. Estimate consumer surplus per trip from a travel-cost demand model that includes transport and lodging costs:

$$P_j = CS_j \quad [\$/\text{trip}].$$

Estimate the forest-purpose share of trip surplus from trip-purpose surveys or a stated-preference/travel-cost counterfactual:

$$\lambda_j = \frac{\max(0, CS_j^{\text{forest purpose}} - CS_j^{\text{other purpose}})}{CS_j^{\text{forest purpose}}}, \quad CS_j^{\text{forest purpose}} > 0.$$

Then value nature tourism as

$$V_{\text{nature tourism}} = \sum_j Q_j P_j \lambda_j,$$

keeping the trip unit consistent with the model and not adding lodging or visitor expenditures to consumer surplus.

Estimated Value

3.3.9 Ecosystem 9: Air Quality Purification (Regulation)

Data. Use the Minnesota FIA forest raster as the spatial base, with FIA species and stand attributes that inform leaf area and pollutant removal. Couple it with canopy or i-Tree pollutant-removal estimates for $PM_{2.5}$, ozone, NO_2, and SO_2, ambient pollution and meteorology, exposed-population data, and pollutant-specific marginal health-damage estimates (for example, from EPA BenMAP). Estimate λ_{ip}, the share of avoided damages attributable to forest removal, accounting for where and when removal changes human exposure.

Valuation Approach. Calculate annual pollutant removal for pixel i and pollutant p from canopy/i-Tree rates:

$$Q_{ip} = A_i r_{ip},$$

where r_{ip} is tons removed per hectare per year; use atmospheric transport/exposure modeling to refine the quantity where available. Estimate marginal health damages avoided per ton using pollutant-specific concentration-response functions, exposed-population data, and health-damage values (for example, BenMAP):

$$P_{ip} = MD_{ip} \quad [\$/\text{ton pollutant}].$$

Estimate the share of modeled damage reduction attributable to eligible family-forest canopy using the ownership overlay and a no-canopy counterfactual:

$$\lambda_{ip} = \frac{\Delta D_{ip}^{\text{eligible forest}}}{\Delta D_{ip}^{\text{all canopy}}},$$

where ΔD is avoided health damage and the denominator is positive. If Q_{ip} already measures only the eligible-forest counterfactual difference, set $\lambda_{ip} = 1$ to avoid applying attribution twice. Then value annual air-quality benefits as

$$V_{\text{air}} = \sum_{i,p} Q_{ip} P_{ip} \lambda_{ip},$$

using consistent units and avoiding counting the same exposure benefit under multiple services.

Estimated Value

3.3.10 Ecosystem 10: Water Quality Regulation from Riparian Buffers

Data. Use the Minnesota FIA forest raster to identify forest cover, species composition, and stand attributes, and combine it with a complete land-use/land-cover raster so InVEST Nutrient Delivery Ratio (NDR) can represent both forest and nonforest sources. Required inputs include a stream network and watershed boundaries, a hydrologically conditioned DEM, a runoff proxy, and an NDR biophysical table with nitrogen and phosphorus loads and retention efficiencies by land-cover class. Delineate riparian buffers using a stated stream-distance/width rule and identify family-forest ownership within them. Run NDR for the observed land cover and a counterfactual in which the targeted riparian forest buffers are replaced by the specified nonforest cover; keep all other inputs and model parameters fixed.

Valuation Approach. For nutrient n and watershed w, calculate the annual nutrient load reduction attributable to riparian buffers from the two NDR scenarios:

$$Q_{nw} = \max\left(0, E_{nw}^{\text{no buffer}} - E_{nw}^{\text{with buffer}}\right),$$

where E_{nw} is NDR-modeled nutrient export in kg/year. Select a pollutant-specific marginal avoided treatment/damage cost or nutrient-credit price:

$$P_{nw} = p_{nw}^{\text{avoided}} \quad [\$/\text{kg nutrient}],$$

using local treatment costs, water-quality damage estimates, or applicable nutrient-credit markets; state the price year and do not add overlapping cost measures. Attribute the modeled buffer benefit to family forests using an NDR scenario that removes only eligible family-forest buffers:

$$\lambda_{nw} = \frac{\Delta E_{nw}^{\text{eligible family forest}}}{\Delta E_{nw}^{\text{all riparian buffers}}}, \quad 0 \le \lambda_{nw} \le 1,$$

where each ΔE is the reduction in exported nutrient relative to its matching no-buffer scenario. If Q_{nw} is already calculated using only eligible family-forest buffers, set $\lambda_{nw} = 1$ to avoid applying attribution twice. Value the annual water-quality regulation service as

$$V_{\text{water}} = \sum_{n,w} Q_{nw} P_{nw} \lambda_{nw}.$$

Use non-nested watershed outlets or otherwise prevent double counting loads, and report nitrogen and phosphorus separately where their prices or damage pathways differ.

Estimated Value

4 What is the Value of Minnesota Family Forests?

- TBD - General breakdown of value by groups (which ecosystem services dominate). Comparison of monetary and non-monetary values across ecosystem services. Do market forces drive the effect? - Want a visual comparision of timber production (bar) against stacked bar of all other ecosystem services. - Time trends. Is there anything to be stated about changes in forest type, ownership patterns, or ecosystem service provision over time? - Heterogeneity in forest types and acerage (size)

5 How Does Forest Value Change under a Warming Climate?

- Projections of climate change impacts on LULC - Implications for ecosystem service provision and forest value - What do we expect for transfer across ownership type (family forest to public or deforestation)? - What is the implication with agricultural expansion, prarie encroachment, increased global competition driving down timber demand (i.e., Eucalyptus spp.)

6 Policy Implications

- Given a fixed MN budget, what policies might we recommend? - How can policy support the sustainable management and conservation of family forests? - Are there trade-offs between different policy options in terms of ecosystem service provision and forest value?

7 Conclusion

Include limitations here. Summarize the principal findings and identify the most important next steps for research, data collection, or decision-making.

References

A Supplementary Appendix

Place detailed methods, assumptions, supporting tables, and additional figures here.

B Glossary of Terminology

- stumpage - stand - site - forest

C Unassessed Ecosystem Services

List and describe ecosystem services that were not assessed in this study, explaining why they were excluded and any potential implications for the overall valuation of Minnesota family forests. Does this lead to underestimation of the total value of Minnesota family forests? - What are the gaps in the literature in what methodologies can be used or what data can be used? - Limitations to measurement.

D Methodological and Data Limitations

Here I want to document the limitations related to the methodology, data sources, assumptions, and any uncertainties that may affect the valuation of Minnesota family forests.

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Conflicts of Interest. The authors declare no conflicts of interest.

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